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$[x,y] = \{(a,b) \in \mathbb{Q} : (a,b) \sim (x,y)\}$

$= \{(a,b) \sim (x,y) : ay = bx\}$

ex

$[1,2] = \{(a,b) \in \mathbb{Q} : (a,b) \sim (1,2)\}$

$= \{(a,b) \in \mathbb{Q} : 2a = 1b\}$

$= \{(a,2a) : a \in \mathbb{Z} \ \& \ a \neq 0\}$

ex

a) $[2,3] = \{(a,b) \in \mathbb{Q} : 3a = 2b\}$

b) $[-4,-6] = \{(a,b) \in \mathbb{Q} : -6a = -4b\}$

$-\cancel{2} \cdot 3a = -\cancel{2} \cdot 2b$

$3a = 2b$

Def. 4

Def. the rational #s to be

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Def. 4

Define the rational #s to be

quotient

$$\mathbb{Q} \overset{\text{def}}{=} \{(x,y) \in \mathbb{Q}\}$$

$$= \{[x,y] : x,y \in \mathbb{Z},\ y \neq 0\}$$

useful-ish lemma

Lemma: $[x,y] = [a,b] \iff xb = ay$

pf sketch

If $[x,y] = [a,b]$, then $(x,y) \in [a,b]$ since $(x,y) \in [x,y]$. Thus $(x,y) \sim (a,b)$ so $xb = ay$.

If $xb = ay$, then $(x,y) \sim (a,b)$, so $(x,y) \in [a,b]$. If $(z,w) \sim (a,b)$, so $(z,w) \in [a,b]$. Thus $[x,y] \in [a,b]$. Other direction done similarly.

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Thm. 2

Let $(x,y) \in \mathbb{Q}$. Then $[x,y] = [0,1] \iff x = 0$.

<u>pf</u>

($\Rightarrow$) If $[x,y] = [0,1]$, then $x \cdot 1 = y \cdot 0$ by lemma.

i.e., $x = 0$.

($\Leftarrow$) Conversely, if $x = 0$, then $x \cdot 1 = 0 = y \cdot 0$.

Thus $[x,y] = [0,1]$ by lemma.

Thm. 3

$[x,y] = [1,1] \iff x = y$

$\therefore 0 \left\{ \frac{x}{y} = \frac{1}{1} = 1 \iff x = y \right\}$

<u>pf</u> → similar to above

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Def. 5

Define the binary operations $(+)$ & $(\cdot)$ on $\mathbb{Q}$ as follows. If $[x,y]$ & $[a,b] \in \mathbb{Q}$,

mult. $[x,y] \cdot [a,b] \overset{\text{def}}{=} [xa, yb]$

add. $[x,y] + [a,b] \overset{\text{def}}{=} \left[\frac{xb+ay}{yb}\right]$

$$\frac{x}{y} + \frac{a}{b} = \frac{xb}{yb} + \frac{ay}{yb} = \frac{xb+ay}{yb}$$

(now 4 well-definedness...)

ex

$[1,2] = [3,6]$ & $[2,3] = [-4,-6]$,

ADDITION

so $[1,2] + [2,3] = [3,6] + [-4,-6]$

$\Rightarrow [3+4, 6] = [7,6]$

$\Rightarrow [3\cdot(-6) + 6(-4), -36]$

$\Rightarrow [-18-24, -36]$

$\Rightarrow [-42, -36]$

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so $[1,2]+[2,3]=[3,6]+[-4,-6]$

$\Rightarrow [3+4,6]=[7,6]$

$\Rightarrow [3\cdot(-6)+6(-4),-36]$

$\Rightarrow [-18-24,-36]$

$\Rightarrow [-42,-36]$

MULTIPLICATION

so $[1,2]\cdot[2,3]=[3,6]\cdot[-4,-6]$,

$\Rightarrow [2,6]=[-12,-36]$

$\Rightarrow 2\cdot(-36)=6\cdot(-12)$ ✓

adding ~~wrong~~:

don't do this...

$[1,2]+[2,3]=[3,6]+[-4,-6]$

$\quad\quad\quad

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Lemma 4 & 5: $+, \cdot$ are well defined

Thm. 6

Additive identity is $[0,1]$

multaplicative identity is $[1,1]$

Thm. 7

$[x,y] = [cx, cy]$ if $c \in \mathbb{Z}$ & $c \neq 0$.

Thm. 8, 9, 10

$+, \cdot$ are commutative, & $[x,y] \cdot [y,x] = [1,1]$ if $x, y \neq 0$.

Thm. 11

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if $x, y \neq 0$.

Thm. 11 $\mathbb{Q}$ is a field.

Every element of $\mathbb{Q}$ other than $[0,1]$ has a multaplicative inverse.

MULT INVERSE

$$\frac{101}{211} \cdot \frac{211}{101} = 1$$

ADD. INVERSE

$$\frac{101}{211} + \left(-\frac{101}{211}\right) = \frac{0}{211} = 0$$

Pf

Let $[x,y] \in \mathbb{Q}$ & $[x,y] \neq [0,1]$.

Thus $x \neq 0$. Hence $[y,x] \in \mathbb{Q}$.

We compute: $[x,y] \cdot [y,x] = [xy, xy] = [1,1]$

$\square$

Def. 6

injective

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Def. 6

The canonical embedding $\mathbb{Z} \overset{\text{injective}}{\hookrightarrow} \mathbb{Q}$ is defined by

$$a \longmapsto [a,1] \quad \text{for every } a \in \mathbb{Z}.$$

This allows us to thinke of $\mathbb{Z}$ as a subset of $\mathbb{Q}$.

Thm. 12

The canonical embedding is injective.

pf

Suppose $a \longmapsto [c,1]$ & $b \longmapsto [c,1]$.

Thus $[a,1] = [c,1]$ & $[b,1] = [c,1]$.

So $[a,1] = [b,1]$. Hence $a \cdot 1 = b \cdot 1 \Rightarrow a = b$.

$\square$

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Suppose $a \longmapsto [c,1]$ & $b \longmapsto [c,1]$.

Thus $[a,1] = [c,1]$ & $[b,1] = [c,1]$.

So $[a,1] = [b,1]$. Hence $a \cdot 1 = b \cdot 1 \Rightarrow a = b$.

$\square$

$+, \cdot, -$ all behave well

$1+2=3 \longmapsto [3,1]$ ✓ yay!

$[1,1] + [2,1] = [1\cdot 1 + 2\cdot 1, 1] = [3,1]$

$[a,1] = a$

$[a,b] - [x,y] \overset{\text{def}}{=} [a,b] + [-x,y]$